Stakeholder-driven municipal air-quality batch system

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Purpose and stakeholders

The product is a batch data service for municipal air-quality work. Analysts study trends and doubtful readings; planners compare locations and policy periods; dashboard developers request recent values. A data steward checks runs, while an authorised officer reviews possible alert cases. The system does not send warnings to citizens.

Evidence and limitations

The UCI Beijing dataset provides 420,768 hourly observations from 12 stations. It supports testing of volume and missing values, but it cannot describe another municipality. The files do not confirm calibration, sensor precision or spatial coverage, and hourly data may not answer every health question.

Database choice

MongoDB stores each station-hour as a flexible document with protected common fields and quality labels. Indexes support time and station queries. PostgreSQL is stronger for a stable relational model, CSV files provide weak concurrent access, and a warehouse would add too much scope.

Plan and success criteria

Implementation will use batches of 1,000, deterministic IDs, insert-only upserts, limited retries, quarantine records and run heartbeats. Success means all source rows are reconciled, no duplicate station-hours appear, replay is harmless, quality results are visible, and four user queries stay below two seconds. Silent replacement of missing data, failed runs or an unapproved alert are treated as failures.

Sources: UCI Machine Learning Repository, Beijing Multi-Site Air Quality, https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data; MongoDB, Data Modeling, https://www.mongodb.com/docs/manual/data-modeling/